IS507

# Predicting death in heart failure patients

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# Motivation and potential impact

## Why we picked it

We wanted a real medical problem where the answer matters, and a dataset small enough to study carefully. It also forces us to deal with leakage, imbalance and small samples, not just chase accuracy.

## Why it could matter

A model that flags higher-risk patients could help clinicians look closer, earlier. It could also show which measurements carry the most signal.

Heart failure carries a high risk of death, so even a modest gain in spotting high-risk patients is worth studying.

# Problem and goal

### Input (X)

12 clinical measurements per patient, or 11 when we drop follow-up time.

### Output (Y)

Did the patient die during follow-up? Yes or no.

### Loss and metrics

Logistic Regression and XGBoost minimize log loss. We judge all models on accuracy, recall, F1 and ROC-AUC.

Goal: 80% accuracy. Follow-up lasts 4 to 285 days, so we predict death during follow-up, not 1, 2 or 3 year survival.

# Related work

### The original records

Ahmad et al. (2017) published the patient records from a heart failure study.

### Chicco and Jurman (2020)

Applied machine learning to this dataset and reported that serum creatinine and ejection fraction alone can predict survival.

### Where we add

We test with and without follow-up time, use repeated cross-validation, and check whether synthetic records help on real held-out patients.

We compare three models under a setup that guards against leakage from the time variable.

# The data

**299 × 13**

299 patients, one per row. 12 features plus the outcome.

**7 + 5**

7 numeric features (age, creatinine, sodium, time) and 5 binary ones (anaemia, sex, smoking).

**32% died**

96 of 299 patients died, so the classes are imbalanced.

Source: UCI Heart Failure Clinical Records (Chicco and Jurman, 2020). We found no missing values and no duplicate rows.

# The baseline

**0.678**

accuracy from guessing "survived" for every patient

**0**

deaths caught. Recall is 0 and ROC-AUC is 0.50

**12 pts**

gap between this baseline and our 80% goal

Why this baseline: it is the simplest possible predictor, so any useful model has to beat it. Accuracy alone hides missed deaths, so we also compare recall, F1 and ROC-AUC.

# The variable called time

### What it is

Follow-up days. It ends when a patient dies or leaves the study, so a doctor would not know it at prediction time.

**6 to 11 pts**

drop in accuracy when time is removed

**10 to 17 pts**

drop in ROC-AUC when time is removed

Time probably makes the models look better than they are, so every model runs both ways and the with-time results are an upper bound.

# How we train and evaluate

### 1. Split once

Stratified 80/20: 239 training and 60 test patients. The test set stays untouched until the end.

### 2. Cross-validate

5-fold cross-validation repeated 10 times, on training data only, since one validation set from 299 patients is noisy.

### 3. Add synthetic data

1,000 artificial records made from training data only, never used for testing.

Every model runs with and without time, and we compare real-only against real plus artificial training on the same held-out real patients.

# The three models

### Logistic Regression

Simple and readable. Features are standardized, and the coefficients show which variables push risk up or down.

### Random Forest

Many decision trees voting together. Handles non-linear effects and gives importance scores for each feature.

### XGBoost

Trees built one after another, each correcting the last one's mistakes. Often strong on small tables like ours.

All three use class weights for the imbalance, and we tune them by cross-validation on the training set only.

# Preliminary results

| Features | Model | Accuracy | Recall | AUC |
|---|---|---|---|---|
| With time | Logistic Reg. | 0.805 | 0.782 | 0.882 |
| With time | Random Forest | 0.842 | 0.777 | 0.916 |
| With time | XGBoost | 0.837 | 0.724 | 0.912 |
| No time | Logistic Reg. | 0.741 | 0.710 | 0.786 |
| No time | Random Forest | 0.747 | 0.614 | 0.775 |
| No time | XGBoost | 0.726 | 0.598 | 0.742 |

Cross-validation, training set only, no tuning yet. Baseline accuracy: 0.678.

## Notebook so far

- Data checked and split
- Baseline built
- All six runs done with repeated cross-validation

Only the with-time runs reach 80%. Without time, our best accuracy is 0.747.

# What we expect to find

### With time

The 80% goal is reachable, but it is an upper bound.

### Without time

Mid 70s accuracy, above baseline but short of 80%. The models catch roughly 60 to 71% of deaths.

### Which model wins

Probably no clear winner. The three are within about 2 points, and folds vary by 4 to 6 points.

Tuning and artificial records may help a little. We will only count that if it shows up on real held-out patients.

# Challenges

### Already seen

- Only 299 patients from one study
- 68/32 class imbalance
- Time leaks outcome information
- Mixed numeric and binary features

### Expected ahead

- Noisy scores, 4 to 6 points between folds
- Synthetic records are not new patients
- Overfitting while tuning on tiny data
- Models too close to call

Next: tune the models, compare real-only against real plus synthetic, check feature importance, then run the 60-patient test set once.